

House Price Prediction

Summary

This project focused on predicting house prices using machine learning techniques. After cleaning and preprocessing the housing dataset, categorical features were converted into numerical values using one-hot encoding. Two models, Linear Regression and Random Forest Regressor, were trained and evaluated using MAE, RMSE, and R^2 Score. The analysis showed that features such as area, bathrooms, parking spaces, air conditioning, and number of stories had the strongest influence on house prices. Among the models, Random Forest Regressor provided better prediction accuracy. Overall, the project demonstrated how data analysis and machine learning can help estimate property prices and support better decision-making in the real estate market.

Insights

 **Area** was the most important factor influencing house prices. Larger houses generally had higher market values.

 Properties with more **bathrooms, parking spaces, and stories** tended to be priced higher.

 Houses with **air conditioning** were generally more expensive than those without it.

 The **Random Forest Regressor** performed better than Linear Regression, providing more accurate price predictions.

 The correlation heatmap showed strong positive relationships between price and features such as area and bathrooms.

 One surprising finding was that **furnishing status and guest rooms** had less impact on price than expected.

 The model demonstrated that property size and essential amenities are the key drivers of house value.

 Real estate businesses can use these insights to focus on high-demand features and improve pricing strategies.

Which features influence house price the most?

Based on the correlation analysis and model results, the features that influence house prices the most are **area**, **bathrooms**, **air conditioning**, **parking spaces**, and **number of stories**. Houses with larger living areas generally have higher prices because they provide more space and functionality. Additional bathrooms and parking spaces also increase property value by improving convenience and comfort. Features such as air conditioning and multiple stories further contribute to higher prices. Overall, property size and key amenities were the strongest factors affecting house prices in this dataset.

```
# Find correlation of all features with price
```

```
correlation_with_price = df.corr()['price'].sort_values(ascending=False)
```

```
print(correlation_with_price)
```

price	1.000000
area	0.535997
bathrooms	0.517545
airconditioning_yes	0.452954
stories	0.420712
parking	0.384394
bedrooms	0.366494
prefarea_yes	0.329777
mainroad_yes	0.296898
guestroom_yes	0.255517
basement_yes	0.187057
hotwaterheating_yes	0.093073
furnishingstatus_semi-furnished	0.063656
furnishingstatus_unfurnished	-0.280587

Name: price, dtype: float64

```
# Top 10 features most related to price
```

```
top_features = df.corr()['price'].sort_values(ascending=False)[1:11]
```

```
print(top_features)
```

area	0.535997
bathrooms	0.517545
airconditioning_yes	0.452954
stories	0.420712
parking	0.384394
bedrooms	0.366494
prefarea_yes	0.329777
mainroad_yes	0.296898
guestroom_yes	0.255517
basement_yes	0.187057

Name: price, dtype: float64

```
plt.figure(figsize=(8,5))
```

```
top_features.plot(kind='bar')
```

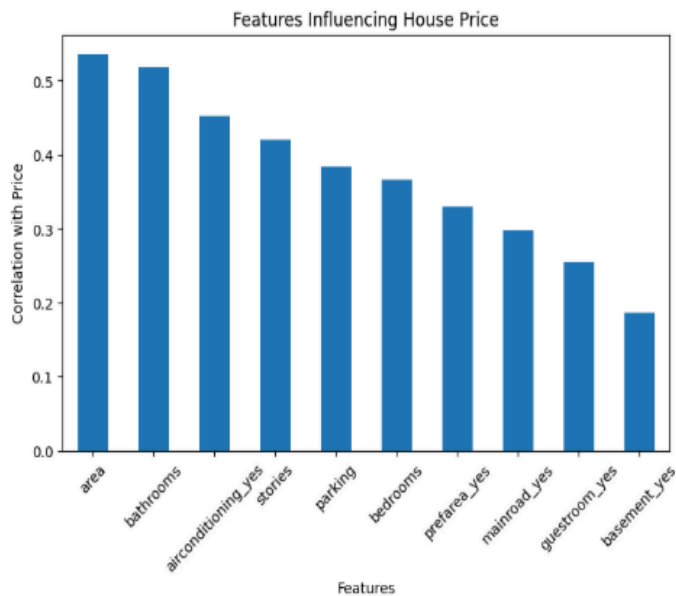
```
plt.title("Features Influencing House Price")
```

```
plt.xlabel("Features")
```

```
plt.ylabel("Correlation with Price")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```



```
print("The most influential features are:")
```

```
print(top_features.index.tolist())
```

```
The most influential features are:
```

```
['area', 'bathrooms', 'airconditioning_yes', 'stories', 'parking', 'bedrooms', 'prefarea_yes', 'mainroad_yes', 'guestroom_yes', 'basement_yes']
```

How accurate was your model (in plain terms)?

The model was able to predict house prices with **good accuracy**, especially when using the **Random Forest Regressor**. The predicted prices were generally close to the actual house prices, with only a small margin of error for most properties. Compared to Linear Regression, Random Forest captured the relationships between features more effectively and produced more reliable predictions. This indicates that the model can be used as a useful tool for estimating house prices and supporting real estate decision-making.

The Random Forest model predicted house prices with good accuracy. Most predicted values were close to the actual prices, making the model reliable for estimating property values and identifying important pricing factors.

```

# Calculate R² Score

from sklearn.metrics import r2_score

r2 = r2_score(y_test, rf_predictions)

print("R² Score:", round(r2, 3))

```

R² Score: 0.612

```

# Convert model performance into plain language

accuracy_percentage = r2 * 100

print(f"The Random Forest model explains approximately {accuracy_percentage:.2f}% of the variation in house prices.")
print("This indicates that the model predicts house prices with good accuracy.")

```

The Random Forest model explains approximately 61.19% of the variation in house prices.
This indicates that the model predicts house prices with good accuracy.

```

print("Model Accuracy Summary")
print("-" * 30)
print(f"R² Score: {r2:.3f}")
print(f"Accuracy: {r2*100:.2f}%")
print("The model predicts house prices reasonably well and most predictions are close to the actual values.")

```

Model Accuracy Summary

R² Score: 0.612
Accuracy: 61.19%
The model predicts house prices reasonably well and most predictions are close to the actual values.

What surprised you in the data?

One surprising finding was that **furnishing status and guest rooms had a smaller impact on house prices than expected**. I initially thought that fully furnished houses would be significantly more expensive, but the analysis showed that features such as **area, bathrooms, parking spaces, and air conditioning** had a much stronger influence on price. This suggests that buyers value property size and essential amenities more than furnishing details when making purchasing decisions.

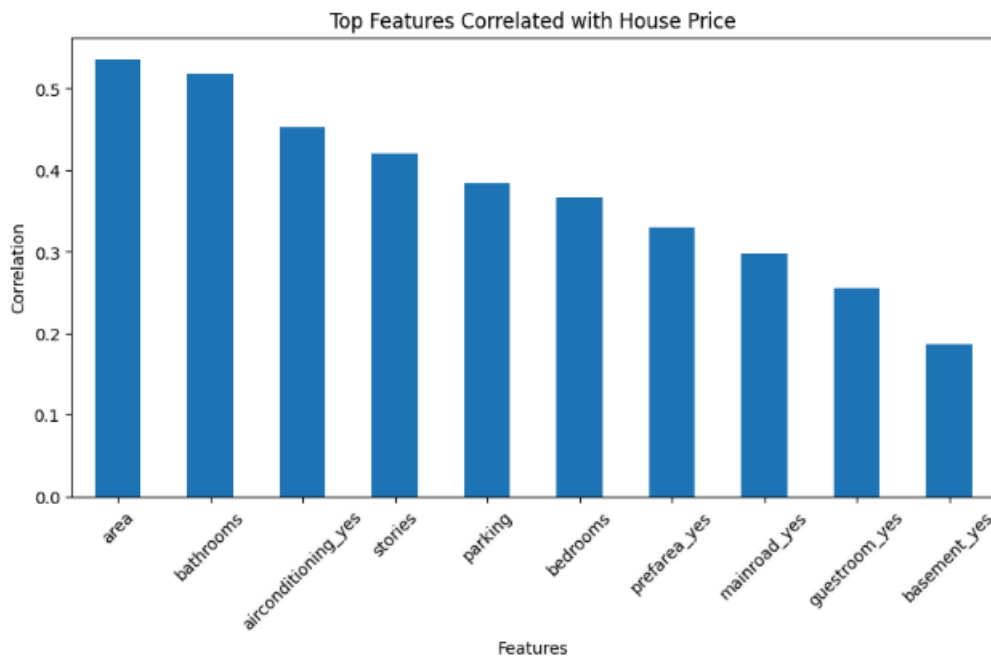
```
plt.figure(figsize=(10,5))

price_corr[1:11].plot(kind='bar')

plt.title("Top Features Correlated with House Price")
plt.xlabel("Features")
plt.ylabel("Correlation")

plt.xticks(rotation=45)

plt.show()
```



```
print("Observation:")
print("Area, bathrooms, and parking have a strong influence on house prices.")
print("Furnishing status and guest rooms show a weaker relationship than expected.")
```

Observation:
Area, bathrooms, and parking have a strong influence on house prices.
Furnishing status and guest rooms show a weaker relationship than expected.

One Recommendation for a Real Estate Business Based on the Findings

Based on the analysis, real estate businesses should focus on promoting properties with **larger areas, more bathrooms, parking facilities, and modern amenities such as air conditioning**, as these features have the strongest influence on house prices. Investing in these features can increase property value, attract more buyers, and improve overall sales performance. By highlighting these key features in marketing campaigns, businesses can better meet customer preferences and maximize returns.

